

LetsDefend Alert - Event ID 139

Description

```
EventID: 139
Event Time: Apr, 20, 2023, 09:42 AM
Rule: SOC189 - VBScript Suspicious Behavior Detected
Level: Incident Responder
Hostname: David
IP Address: 172.16.17.31
Related Binary: Purchase_Order.xls.vbs
Binary Path: C:\Users\LetsDefend\Downloads\Purchase_Order\Purchase_Order.xls.vbs
Binary MD5: 8FAF36EDFAE1EC0E8ECCD3C562C03903
Trigger Reason: VBScript attempting to access sensitive system resources or files,
such as the Windows Registry or system files, that are not related to its expected
functionality.
Device Action: Allowed
L1 Note: When i searched the hash online the file seems it is some variant of wshrat.
And it seems malicious. I'm assigning this alert for further investigations.
```

Investigation

Let's examine the Endpoint Security tab.

Host information

```
Hostname: David
IP Address: 172.16.17.31
Last Login: Apr, 20, 2023, 11:22 AM
```

Browser History

```
2023-04-20 09:40
hxxps[:]//files-ld.s3.us-east-2.amazonaws[.]com/Purchase_Order.zip
```

Terminal History

No records

Processes

The wscript.exe

Execution of the script is confirmed:

```
Event Time: 2023-04-20 09:42:06.918
Process ID: 6640
Image Path: C:\Windows\System32\wscript.exe
Process User: EC2AMAZ-ILGV0IN\LetsDefend
Parent Name: explorer.exe
Parent Path: C:\Windows\explorer.exe
Command Line:
"C:\Windows\System32\WScript.exe"
"C:\Users\LetsDefend\Downloads\Purchase_Order\Purchase_Order.xls.vbs"
```

User: LetsDefend

Parent: explorer.exe (user double-click execution)

Red flags:

- Double extension (the file pretends to be an Excel file `Purchase_Order.xls.vbs` , actually VBScript)
- Execution from `Downloads` directory (often files downloaded by users, frequently used in initial infection)
- Execution via `wscript` (it is a legitimate tool, but often abused to run malicious scripts e.g.: droppers, downloaders, persistence mechanism)

Other observed processes

Task Scheduler (Schedule service)

Used by Windows to run automated tasks (updates, maintenance, scripts, etc.).

```
Parent Path:
C:\Windows\System32\services.exe

Command Line:
C:\Windows\system32\svchost.exe -k netsvcs -p -s Schedule
```

The path and the parent (`services.exe`) look legitimate.

Windows Modules Installer

Installs, modifies, and removes Windows updates.

Handles system components and protected files.

```
Parent Path:
C:\Windows\System32\services.exe

Command Line:
C:\Windows\servicing\TrustedInstaller.exe
```

The path and the parent (`services.exe`) look legitimate.



Tip

The suspicious parents would be e.g.: `cmd.exe` , `powershell.exe`

VirusTotal

Let's examine the suspicious file executed by the user, as observed in the endpoint logs:

`Purchase_Order.xls.vbs`

According to the note the MD5 hash is `8faf36edfae1ec0e8eccd3c562c03903`

<https://www.virustotal.com/gui/file/1c546a6548beda639640ebfbb52abd5f6013c33500172cfccf0e8716c96bb196>

The file was identified as malicious by **35** vendors.

Community Score: **-46**

Note the information found in `Relations` tab on VirusTotal:

DETECTION

DETAILS

RELATIONS

BEHAVIOR

COMMUNITY

14 +

Join our Community

 and enjoy additional community insights and crowdsourced detections, plus an API key to [automate checks](#).

Contacted URLs (1)

Scanned	Detections	Status	URL
2024-12-15	14 / 96	-	http://chongmei33.publicvm.com:7045/is-ready

Contacted Domains (2)

Domain	Detections	Created	Registrar
chongmei33.publicvm.com	18 / 91	2007-07-19	Netdorm, Inc. dba DNSExit.com
publicvm.com	8 / 91	2007-07-19	Netdorm, Inc. dba DNSExit.com

Contacted IP addresses (2)

IP	Detections	Autonomous System	Country
103.47.144.80	1 / 91	205009	PK
46.246.86.2	8 / 91	42708	SE

The file was reported as malicious also by the LetsDefend Threat Intel.

Network Action

We can see several connections to the IP address: `103.47.144.80`

The IP address is identified by VirusTotal as related to the malware.

The IP was not found in neither AbuseIPDB nor LetsDefend Threat Intel.

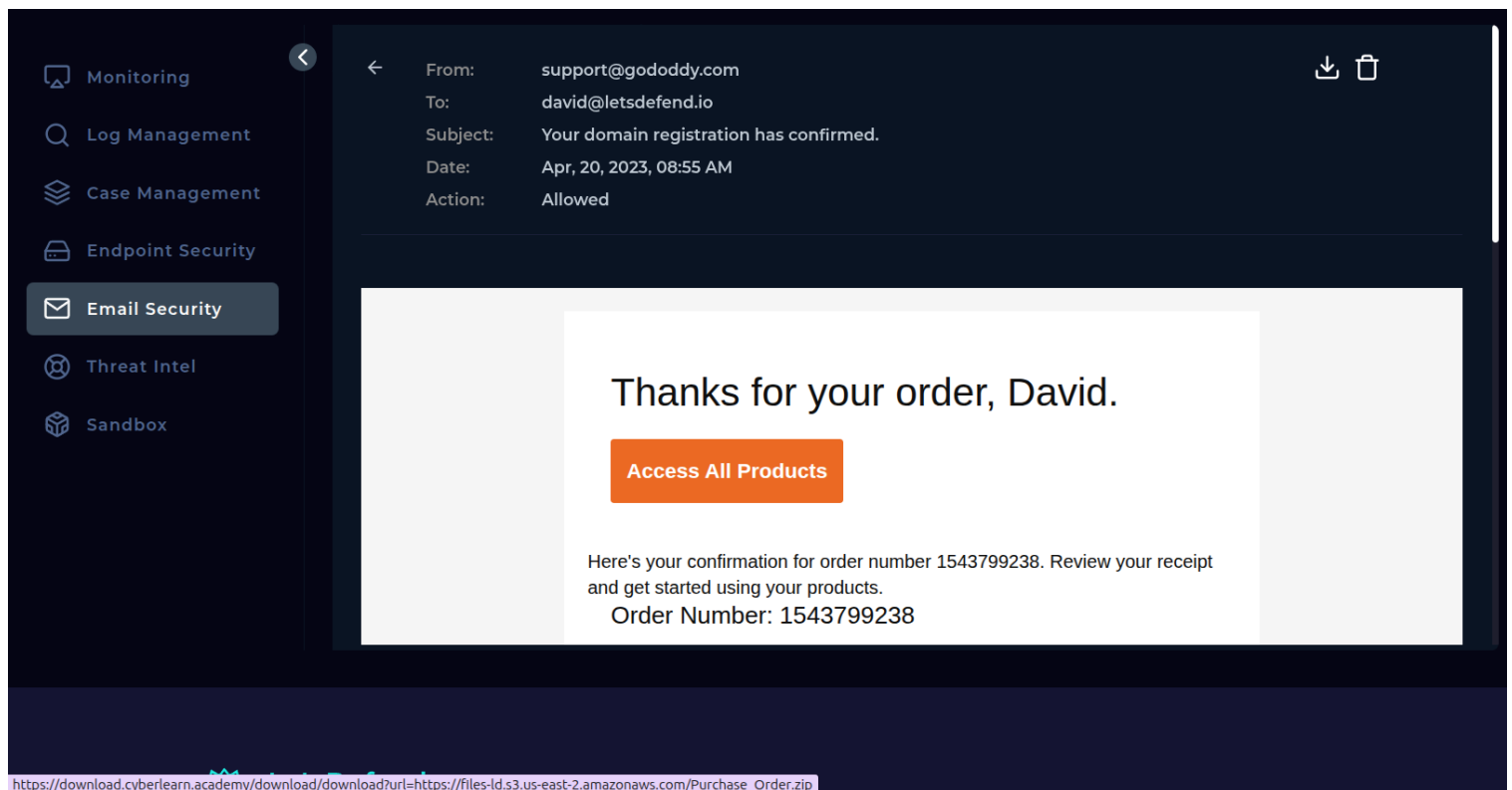
```
2023-04-20 09:42:10.383 103.47.144.80
2023-04-20 09:42:30.283 103.47.144.80
2023-04-20 09:43:32.21 103.47.144.80
2023-04-20 09:44:34.124 103.47.144.80
2023-04-20 09:45:30.427 103.47.144.80
2023-04-20 09:46:32.205 103.47.144.80
2023-04-20 09:47:34.033 103.47.144.80
```

Repeated connections to the same external IP suggest C2 communication.

Initial Access

The user received a phishing email containing a link to a malicious ZIP archive, which was downloaded and executed.

```
From: support@gododdy.com
To: david@letsdefend.io
Subject: Your domain registration has confirmed.
Date: Apr, 20, 2023, 08:55 AM
Action: Allowed
```



The malicious URL `hxxps[:]//files-ld.s3.us-east-2.amazonaws[.]com/Purchase_Order.zip` can be found in a few places in the email body.

Endpoint Analysis

Connect to the endpoint.

Sysmon

Open the Event Viewer and navigate to Sysmon logs:

Application and Services Logs → Microsoft → Windows → Sysmon → Operational

Investigate Sysmon logs

First of all filter in logs around the incident date: Apr, 20, 2023, 09:42 AM

Proof of downloading the malicious file

Filter logs with Event ID 15 - File stream created

We can see the proof of downloading the malicious zip file.

In the log we can find also the origin URL of the file.

The screenshot displays the Windows Event Viewer interface. The top pane shows a list of events filtered by 'Log: Microsoft-Windows-Sysmon/Operational; Source: ; Event ID: 15'. The bottom pane shows the details of the selected event (Event ID 15, Sysmon).

Event Details:

- ProcessId: 4876
- Image: C:\Program Files\Google\Chrome\Application\chrome.exe
- TargetFilename: C:\Users\LetsDefend\Downloads\Purchase_Order.zip;Zone.Identifier
- CreationUtcTime: 2023-04-20 09:40:40.766
- Hash: MD5=1B24CC64AF68FFBEC495AEABDC13896,SHA256=BD4BB18D9BA4D6109CF2A7029AF5B59C1200E2149BC9ACC46DA662138ACBD37,IMPHASH=00000000000000000000000000000000
- Contents: [ZoneTransfer] ZoneId=3 HostUrl=https://files-ld.s3.us-east-2.amazonaws.com/Purchase_Order.zip
- User: EC2AMAZ-ILGVOIN\LetsDefend

Log Properties:

- Log Name: Microsoft-Windows-Sysmon/Operational
- Source: Sysmon
- Event ID: 15
- Level: Information
- User: SYSTEM
- OpCode: Info
- Task Category: File stream created (rule: FileCreateStreamHash)
- Keywords:
- Computer: EC2AMAZ-ILGVOIN

Proofs of persistence

Filter logs with Event ID 13 - Registry value set

Adding a registry key:

The screenshot shows the Windows Event Viewer window with the 'Operational' log selected. The 'Filter Current Log' dialog is open, showing the 'Filter' tab. The 'Logged' range is set to 'From 4/20/2023 12:00:00 AM to 4/20/2023 11:59:59 PM'. The 'Event level' is set to 'Information'. The 'Event logs' are set to 'Microsoft-Windows-Sysmon/Operational'. The 'Includes/Excludes Event IDs' field contains '13'. The 'Task category' is empty. The 'Keywords' field is empty. The 'User' is set to '<All Users>'. The 'Computer(s)' is set to '<All Computers>'. The 'OK' button is visible.

Level	Date and Time	Source	Event ID	Task Category
Information	4/20/2023 9:42:09 AM	Sysmon	13	Registry value set
Information	4/20/2023 9:42:06 AM	Sysmon	13	Registry value set
Information	4/20/2023 9:40:46 AM	Sysmon	13	Registry value set

Event 13, Sysmon

General Details

Registry value set:
RuleName: T1060,RunKey
EventType: SetValue
UtcTime: 2023-04-20 09:42:06.846
ProcessGuid: {a584806d-08eb-6441-0501-000000005c01}
ProcessId: 6640
Image: C:\Windows\System32\WScript.exe
TargetObject: HKU\S-1-5-21-3163960855-2866672989-1813526453-1008\Software\Microsoft\Windows
CurrentVersion\Run\Purchase_Order
Details: wscript.exe //B "C:\Users\LETSDE~1\AppData\Local\Temp\1\Purchase_Order.xls.vbs"
User: EC2AMAZ-ILGVOIN\LetsDefend

Log Name: Microsoft-Windows-Sysmon/Operational
Source: Sysmon
Event ID: 13
Level: Information
User: SYSTEM
OpCode: Info
More Information: [Event Log Online Help](#)

Log Name: Microsoft-Windows-Sysmon/Operational
Source: Sysmon
Event ID: 13
Level: Information
User: SYSTEM
OpCode: Info
More Information: [Event Log Online Help](#)

Filter logs with Event ID 11 - File created

Adding the script to Startup:

The screenshot shows the Windows Event Viewer window with the 'Operational' log selected. The 'Filter Current Log' dialog is open, showing the 'Filter' tab. The 'Logged' range is set to 'From 4/20/2023 12:00:00 AM to 4/20/2023 11:59:59 PM'. The 'Event level' is set to 'Information'. The 'Event logs' are set to 'Microsoft-Windows-Sysmon/Operational'. The 'Includes/Excludes Event IDs' field contains '11'. The 'Task category' is empty. The 'Keywords' field is empty. The 'User' is set to '<All Users>'. The 'Computer(s)' is set to '<All Computers>'. The 'OK' button is visible.

Level	Date and Time	Source	Event ID	Task Category
Information	4/20/2023 9:50:50 AM	Sysmon	11	File created (rule: FileCreate)
Information	4/20/2023 9:50:44 AM	Sysmon	11	File created (rule: FileCreate)
Information	4/20/2023 9:50:39 AM	Sysmon	11	File created (rule: FileCreate)

Event 11, Sysmon

General Details

File created:
RuleName: T1023
UtcTime: 2023-04-20 09:50:39.005
ProcessGuid: {a584806d-08ee-6441-0701-000000005c01}
ProcessId: 7092
Image: C:\Windows\System32\wscript.exe
TargetFilename: C:\Users\LETSDefend\AppData\Roaming\Microsoft\Windows\Start Menu\Programs\Startup
Purchase_Order.xls.vbs
CreationUtcTime: 2023-04-20 09:42:06.846
User: EC2AMAZ-ILGVOIN\LetsDefend

Log Name: Microsoft-Windows-Sysmon/Operational
Source: Sysmon
Event ID: 11
Level: Information
User: SYSTEM
OpCode: Info
More Information: [Event Log Online Help](#)

Log Name: Microsoft-Windows-Sysmon/Operational
Source: Sysmon
Event ID: 11
Level: Information
User: SYSTEM
OpCode: Info
More Information: [Event Log Online Help](#)

These events indicate persistence via registry modification and startup folder placement.

Proof of communication with external malicious resources

Filter logs with Event ID 22 - Dns query

Event Viewer

File Action View Help

Operational

Number of events: 2,492

Filtered Log: Microsoft-Windows-Sysmon/Operational; Source: ; Event ID: 22

Date Range: From 4/20/2023 12:00:00 AM to

Level	Date and Time	Source	Event ID	Task Category
Information	4/20/2023 9:42:11 AM	Sysmon	22	Dns query (rule: DnsQuery)
Information	4/20/2023 9:42:09 AM	Sysmon	22	Dns query (rule: DnsQuery)
Information	4/20/2023 9:41:22 AM	Sysmon	22	Dns query (rule: DnsQuery)

Event 22, Sysmon

General

Details

Dns query:

RuleName: -

UtcTime: 2023-04-20 09:42:10.115

ProcessGuid: {a584806d-08ee-6441-0701-000000005c01}

ProcessId: 7092

QueryName: chongmei33.publicvm.com

QueryStatus: 0

QueryResults: ::ffff:103.47.144.80;

Image: C:\Windows\System32\wscript.exe

User: EC2AMAZ-ILGVOIN\LetsDefend

Log Name:

Microsoft-Windows-Sysmon/Operational

Source:

Sysmon

Logged:

4/20/2023 9:42:11 AM

Event ID:

22

Task Category:

Dns query (rule: DnsQuery)

Level:

Information

Keywords:

User:

SYSTEM

Computer:

EC2AMAZ-ILGVOIN

OpCode:

Info

More Information:

[Event Log Online Help](#)

Actions

Operational

Open Saved Log...

Create Custom View...

Import Custom View...

Clear Log...

Filter Current Log...

Clear Filter

Properties

Disable Log

Find...

Save Filtered Log File As...

Attach a Task To this Log...

Save Filter to Custom View...

View

Refresh

Help

Event 22, Sysmon

Event Properties

Attach Task To This Event...

Copy

Save Selected Events...

Refresh

Help

MITRE ATT&CK mapping

Initial Access	Execution	Persistence	Privilege Escalation	Stealth	Defense Impairment	Credential Access	Discovery
11 techniques	14 techniques	20 techniques	13 techniques	28 techniques	12 techniques	16 techniques	29 techniques
- Phishing Application - External Remote Services - Hardware Additions - Phishing - Replication Through Removable Media - Supply Chain Compromise - Trusted Relationship - Valid Accounts - Wi-Fi Networks	- Scripting Interpreter - PowerShell - Python - Visual Basic - Windows Command Shell - Exploitation for Client Execution - Hijack Execution Flow - Input Injection - Inter-Process Communication - Native API - Scheduled Task/Job - Shared Modules - Software Deployment Tools - System Services - Trusted Developer Utilities Proxy Execution - User Execution - Windows Management Instrumentation	- Authentication Package - LSASS Driver - Port Monitors - Print Processors - Registry Run Keys / Startup Folder - Security Support Provider - Shortcut Modification - Time Providers - Winlogon Helper DLL - Boot or Logon Autostart Execution - Inter-Process Initialization Scripts - Compromise Host Software Binary - Create Account - Create or Modify System Process - Event Triggered Execution - Exclusive Control - External Remote Services - Modify Authentication Process	- Manipulation - Account Manipulation - Boot or Logon Autostart Execution - Boot or Logon Initialization Scripts - Create or Modify System Process - Domain or Tenant Policy Modification - Escape to Host - Event Triggered Execution - Exploitation for Privilege Escalation - Process Injection - Scheduled Task/Job - Valid Accounts - Deobfuscate/Decode Files or Information - Direct Volume Access - Execution Guardrails - Exploitation for Stealth - Hide Artifacts - Hijack Execution Flow - Indicator Removal - Indirect Command Execution - Masquerading - Obfuscated Files or Information - Pre-OS Boot - Process Injection - Reflective Code Loading - Rootkit	- Browser Fingerprint - Double File Extension - Invalid Code Signature - Masquerade Account Name - Masquerade File Type - Masquerade Task or Service - Match Legitimate Resource Name or Location - Rename Legitimate Utilities - Right-to-Left Override	- Domain or Tenant Policy Modification - Downgrade Attack - Exploitation for Defense Impairment - File and Directory Permissions Modification - Modify Authentication Process - Modify Registry - Prevent Command History Logging - Rogue Domain Controller - Safe Mode Boot - Subvert Trust Controls	- from Password Stores - Exploitation for Credential Access - Forced Authentication - Forge Web Credentials - Input Capture - Modify Authentication Process - Multi-Factor Authentication Interception - Network Authentication Request Generation - Network Sniffing - OS Credential Dumping - Steal or Forge Kerberos Tickets - Steal Web Session Cookie - Unsecured Credentials	- Debugger Evasion - Device Driver Discovery - Domain Trust Discovery - File and Directory Discovery - Group Policy Discovery - Local Storage Discovery - Log Enumeration - Network Service Discovery - Network Share Discovery - Network Sniffing - Password Policy Discovery - Peripheral Device Discovery - Permission Groups Discovery - Process Discovery - Query Registry - Remote System Discovery - Software Discovery - System Information Discovery - System Location

Find More Writeups Here

Info

<https://robjar101.gitlab.io/secstash/>